Topic: IoT-Based Smart Parking System (COMP4436-25-P10)

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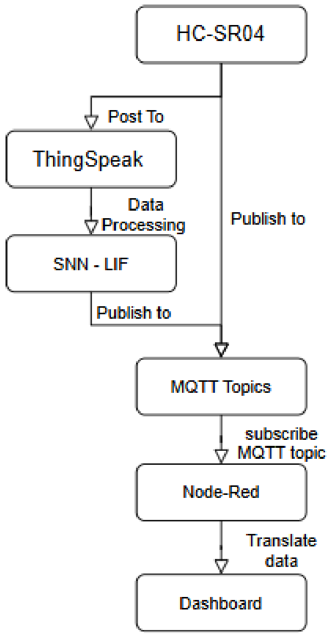
Introduction: Car Park traffic is longtime difficult for management companies. The accurate forecasting and instant information can provide extra revenue for dynamic pricing. For traditional statistic methods may require many pre-processing techniques and related domain knowledge, The deep learning method able to investigate the hidden pattern from the collection of the data thereby able to establish and correct and cheap prediction to increase enterprise turnover.

Data Collection: Since the smart parking system make predictions based on 7-days historical parking data, The ultrasonic sensor (HC-SR04) property of object distance measurement can be implemented for car detection. The sensor will obtain the distance between object and sensor. In this project, 3 ultrasonic sensor setups beneath the parking location, the existing car will consider while object occur distance between 5m to 50m and polling each second of the microcontroller, accumulating for 30 times each minute of detected of car will indicate no room of the parking place.



Data Processing: The Received Data from the sensor will be sent through serial to computer. The computer will publish the data to Node Red server through MQTT and send requests to the Thingspeak Cloud. The 7 days of historical data will be acquired from Thingspeak Cloud. The hour and date of week as X input to SNN (LIF neuron model), to predict 3 parking occupy rate (as y).

Output: The estimation records will produce various reports at node red terminal with the MQTT, the first page of node red will be used predicting occupy rate by hourly, by interval or by holiday each day to suggest the price raising or reducing short-term. The second page will show the predicted occupy rate by hourly, by weekly position to suggest the adjustment of long-term pricing.

datetime (update on 1mins later)	**2025-04-17 01:42:14**
parking space1 is available	**False**
parking space2 is available	**False**
parking space3 is available	**False**

=== Peak Usage Times ===

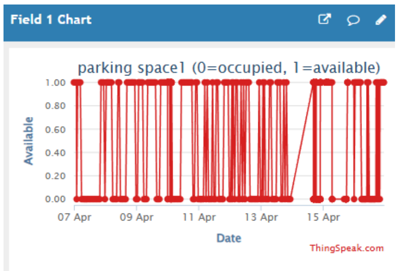
Hour 0400-0500: (Total occupancy rate: 66.67%)
Hour 1000-1100: (Total occupancy rate: 66.67%)
Hour 2100-2200: (Total occupancy rate: 65.1%)

=== Recurring Occupancy Patterns ===

Average WeekDay Occupancy: 48.15%
Average WeekEnd Occupancy: 62.15%

=== Dynamic Pricing Recommendations ===

Hour	Occupancy Rate (%)	Recommended Price (HKD)	Strategy
0000-0100	32.0	17.0	Discount (Low Demand)
0100-0200	43.8	20.0	Maintain (Moderate Demand)
0200-0300	56.3	20.0	Maintain (Moderate Demand)
0300-0400	41.2	20.0	Maintain (Moderate Demand)
0400-0500	66.7	22.0	Increase (High Demand)
0500-0600	37.5	17.0	Discount (Low Demand)
0600-0700	61.9	22.0	Increase (High Demand)
0700-0800	30.6	17.0	Discount (Low Demand)
0800-0900	50.0	20.0	Maintain (Moderate Demand)
0900-1000	61.1	22.0	Increase (High Demand)
1000-1100	66.7	22.0	Increase (High Demand)
1100-1200	50.0	20.0	Maintain (Moderate Demand)
1200-1300	55.6	20.0	Maintain (Moderate Demand)
1300-1400	50.0	20.0	Maintain (Moderate Demand)
1400-1500	38.9	17.0	Discount (Low Demand)
1500-1600	33.3	17.0	Discount (Low Demand)
1600-1700	63.2	22.0	Increase (High Demand)
1700-1800	64.9	22.0	Increase (High Demand)
1800-1900	42.2	20.0	Maintain (Moderate Demand)
1900-2000	55.3	20.0	Maintain (Moderate Demand)
2000-2100	61.9	22.0	Increase (High Demand)
2100-2200	65.1	22.0	Increase (High Demand)
2200-2300	60.2	22.0	Increase (High Demand)
2300-0000	38.9	17.0	Discount (Low Demand)



=== Recommendations ===

Recommendation	Details
Recommendations DateTime	2025-04-17 01:37:09
Recommended Pricing	Should be Higher
Reservation	More_Space

Code

Github Repository :[715yeung/comp4436_AIOT_GP_Group12]

Thingspeak:https://thingspeak.mathworks.com/channels/2909595

=== Daily Parking Utilization Trends (last 7 day & Today) ===

Record Date	Space 1 Occupancy Rate	Space 2 Occupancy Rate	Space 3 Occupancy Rate	Avg Occupancy Rate
2025-04-10	20.25	34.18	39.24	31.22
2025-04-11	64.00	28.00	56.00	49.33
2025-04-12	60.00	64.00	56.00	60.00
2025-04-13	64.00	64.00	64.00	64.00
2025-04-14	74.74	25.26	83.51	61.17
2025-04-15	22.35	80.00	63.53	55.29
2025-04-16	51.02	63.27	44.90	53.06